Cyber Threat Intelligence

Data Collection through Automated OSINT

Tool: Maltego

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1. Introduction

Open Source Intelligence (OSINT) refers to the collection and analysis of data gathered from publicly available sources. Maltego is one of the most widely used OSINT tools that offers powerful data mining and visualization features to investigate digital entities such as people, domains, emails, phone numbers, and more. This report outlines the practical use of Maltego for digital investigations, emphasizing its functionality and real-world applications.

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2. Tool Description

Maltego, developed by Paterva, is a data analysis and visualization platform that enables the mapping of relationships between individuals, groups, websites, and infrastructure components. Using a node-based graph system and a wide variety of OSINT transforms, Maltego uncovers links and data points across various online sources.

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3. Features of Maltego

- Graphical interface for intuitive investigation
- Over 60 pre-installed entities like domains, people, IPs, organizations
- Access to hundreds of transforms via the Transform Hub
- Customizable transform creation and automation
- Integration with APIs, databases, and paid intelligence feeds
- Support for exporting graphs and findings to reports

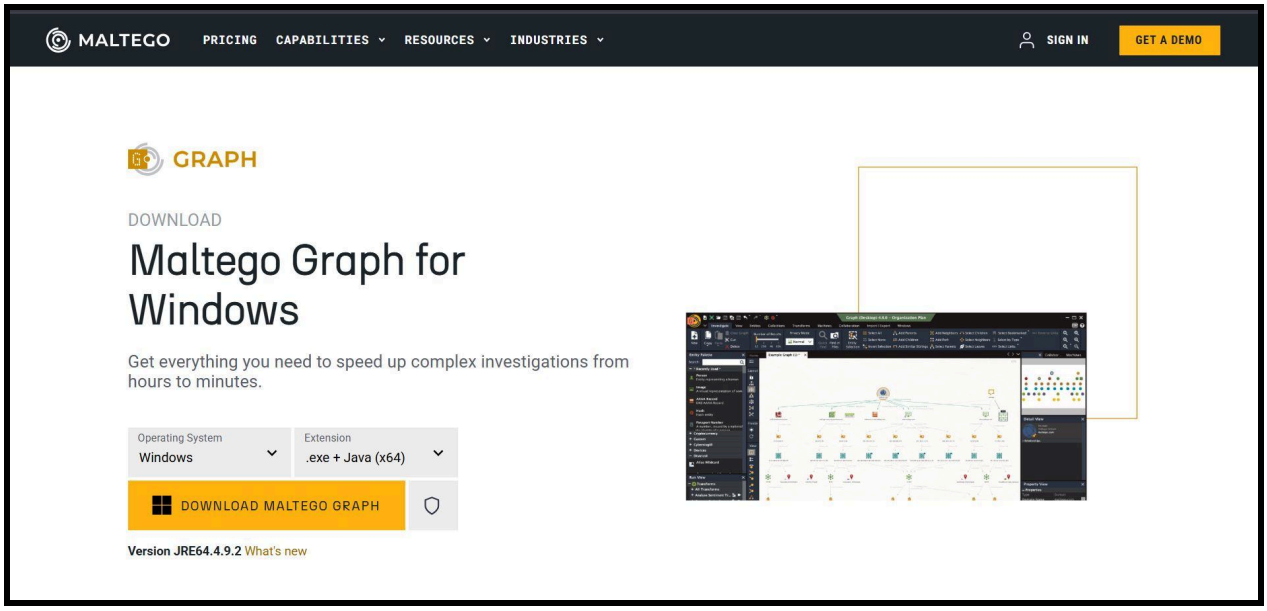
4. Why Use Maltego?

- To visualize hidden relationships between digital entities
- Useful in cybersecurity, digital forensics, and law enforcement
- Facilitates quick identification of attack surfaces and vulnerabilities
- Enhances investigative workflows through automation and visualization

5. How to Use Maltego (Step-by-Step)

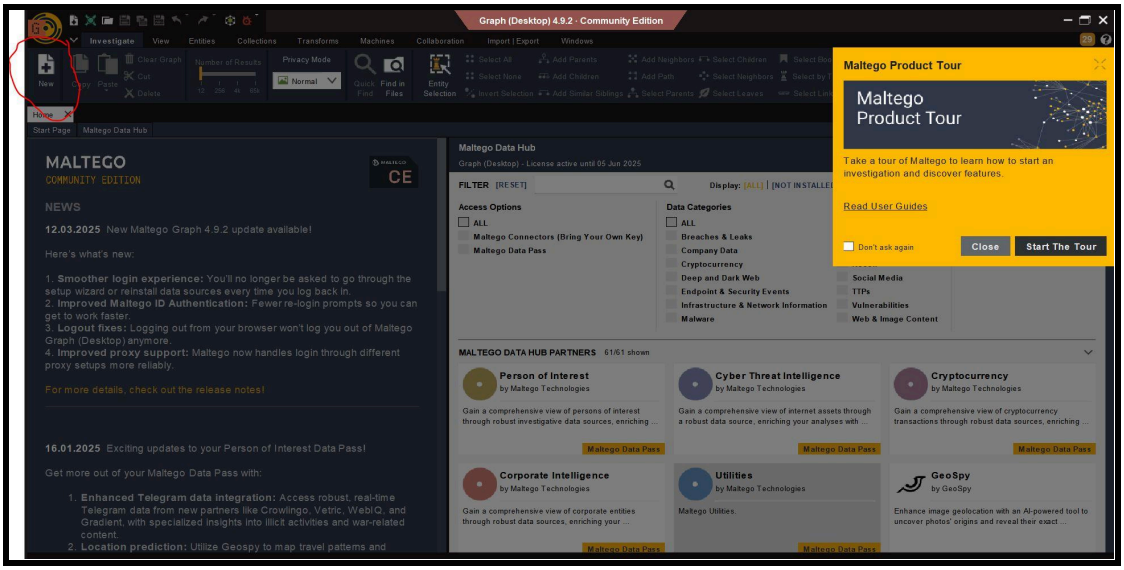
Step 1: Installation and Account Setup

- Download Maltego from the official website: https://www.maltego.com/downloads
- Install the software based on your OS
- Register and log in to the Community or Commercial Edition
- Screenshot:



Step 2: Creating a New Investigation

- Open Maltego and select "New Graph"

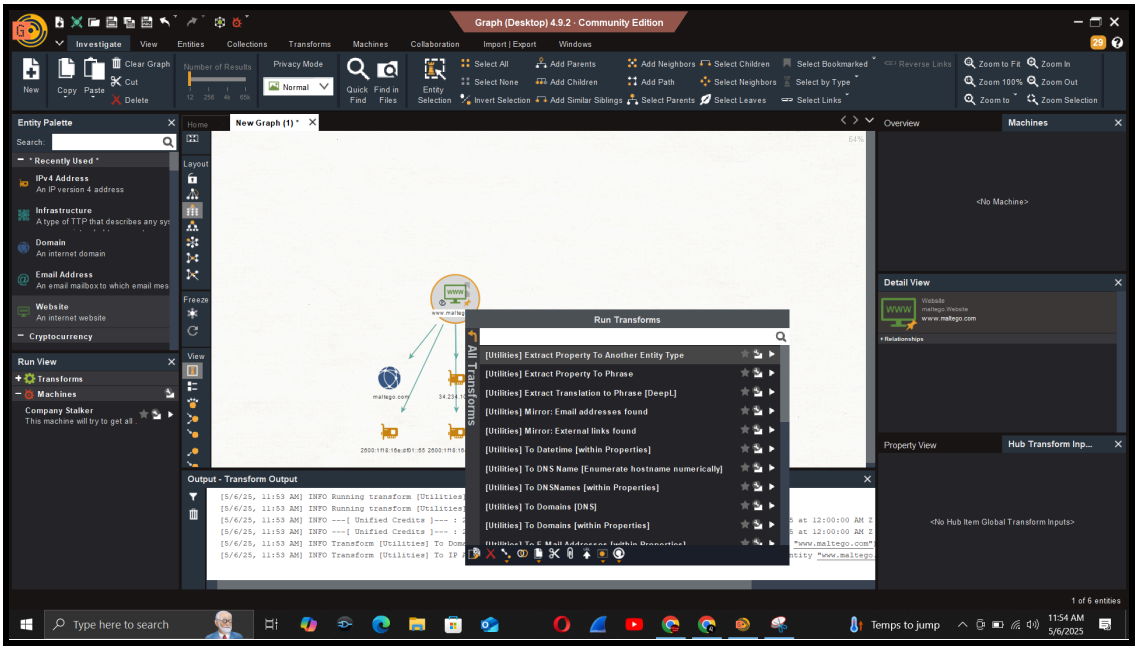


- Add entities like Domain, Person, or Website from the sidebar

Step 3: Running Transforms

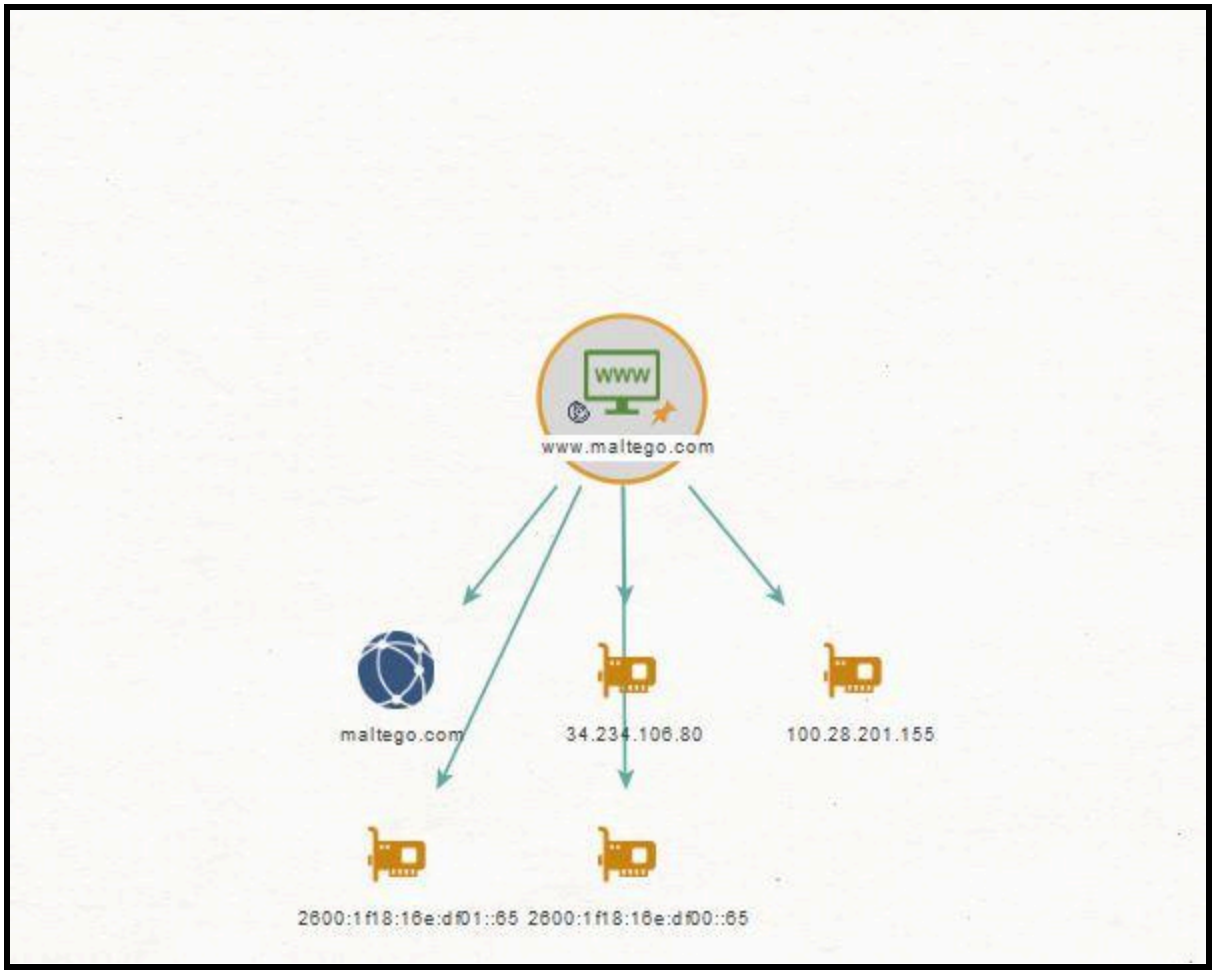
- Right-click on an entity (e.g., domain: `example.com`)
- Choose transforms like:
    - To DNS Name
    - To IP Address
    - To WHOIS Record

    Screenshot



Step 4: Analyzing the Graph

- Review connections and data points created by each transform
- Use layout options to organize the graph visually



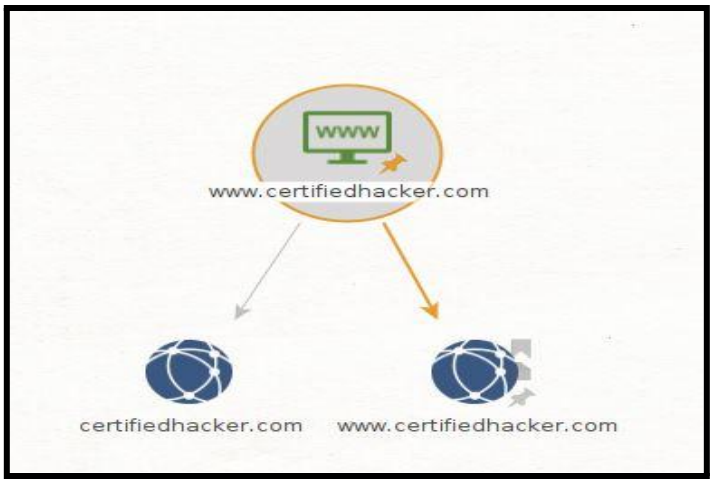
Step 5: Exporting Results

- Save the investigation as a `.mtgl` file
- Export graph as PNG, JPG, or PDF for reports
- Export findings to CSV or built-in report templates

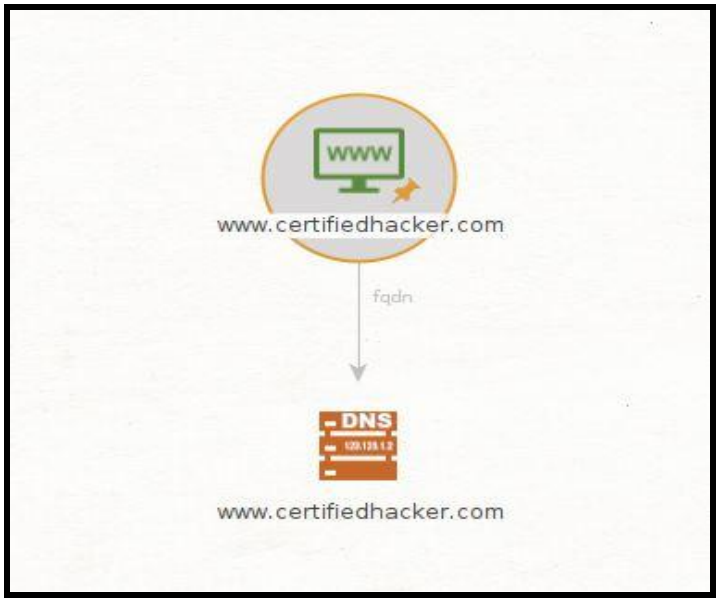
6. Case Studies: Domain Intelligence

Target 1: www.certifiedhacker.com

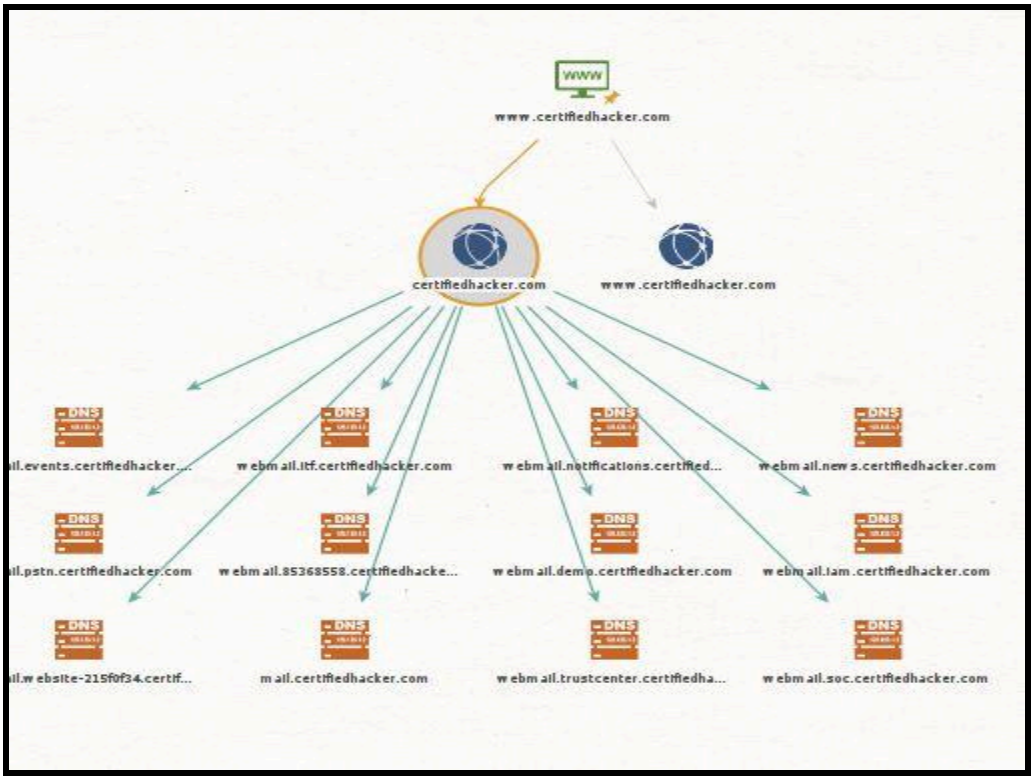
- Discovered associated DNS names



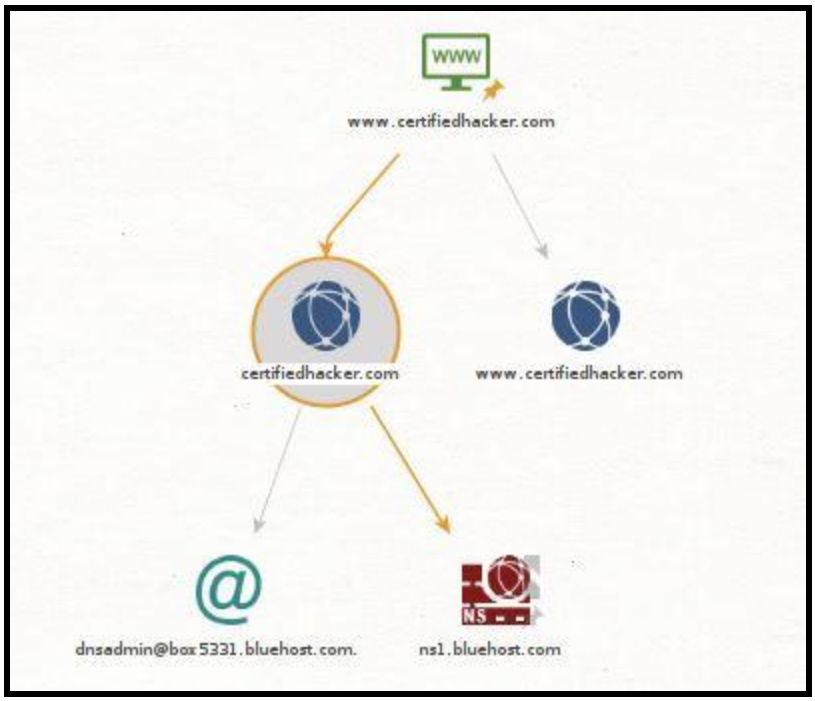
- Found related domains hosted



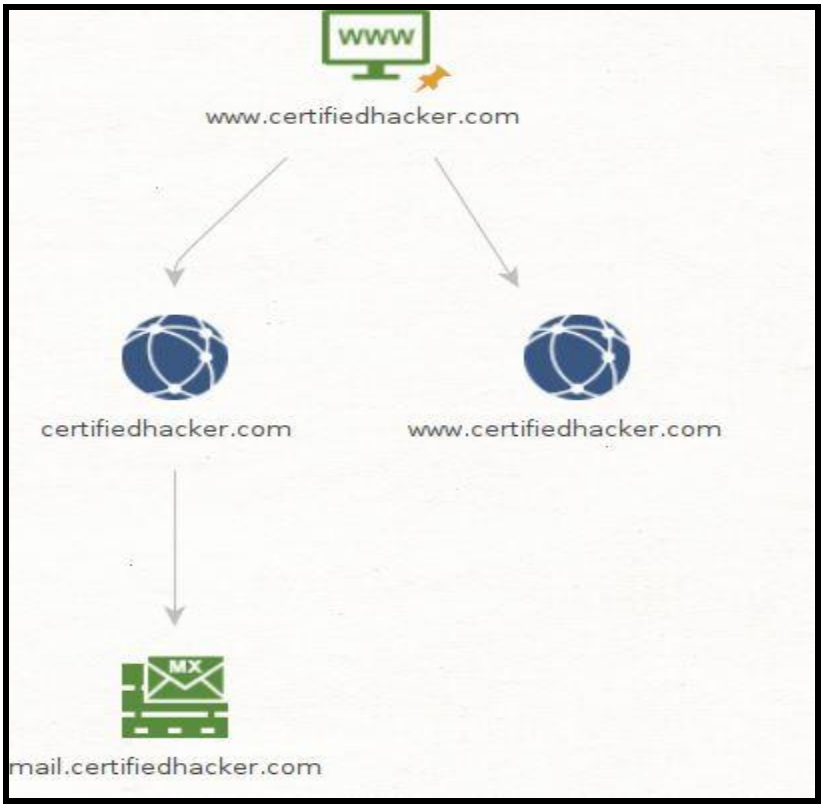
- ***For Subdomain***



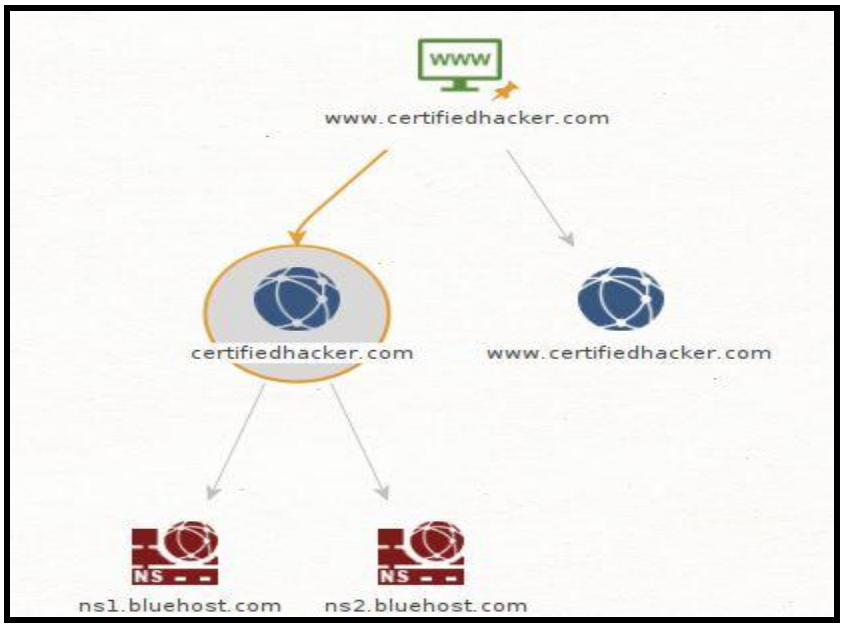
- ***To DNS Name (Start of Authority)***



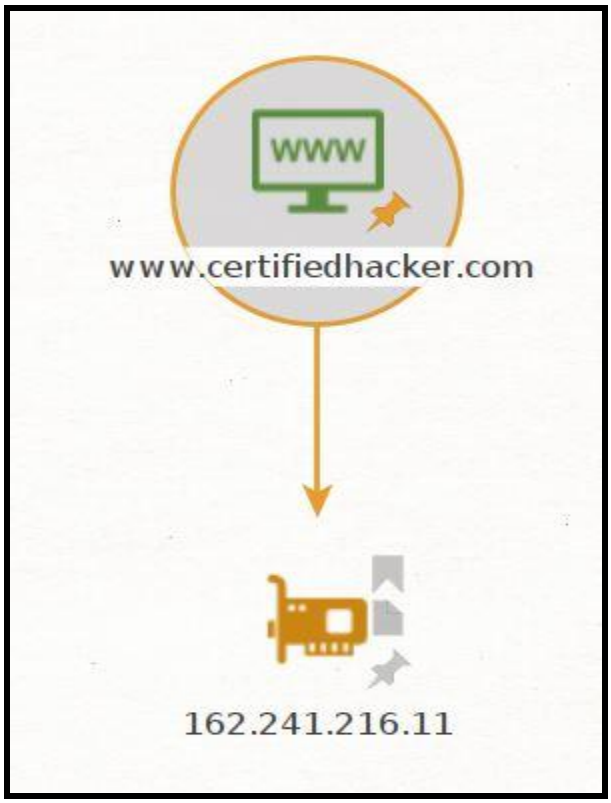
- ***To DNS Name (Mail Server)***



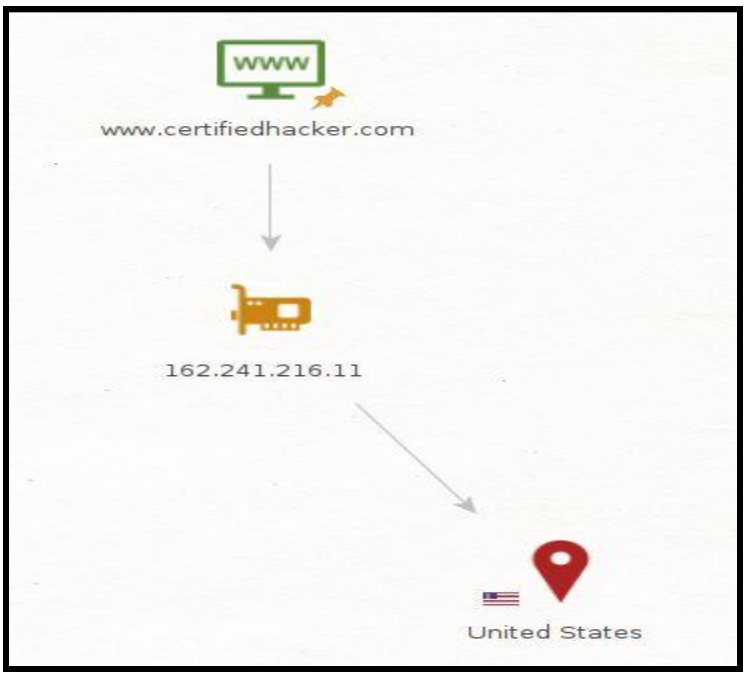
- ***To DNS Name (Name Server)***



- ***To Ip address DNS***

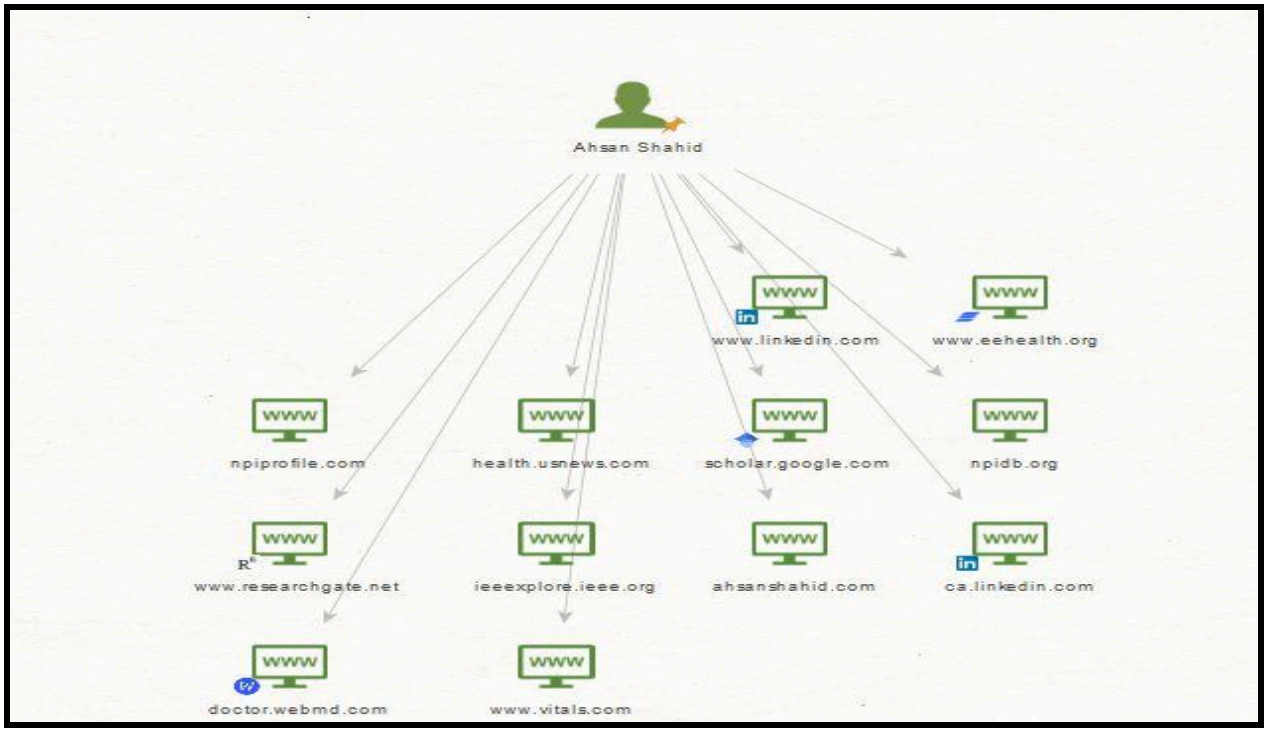


1. ***To Location***



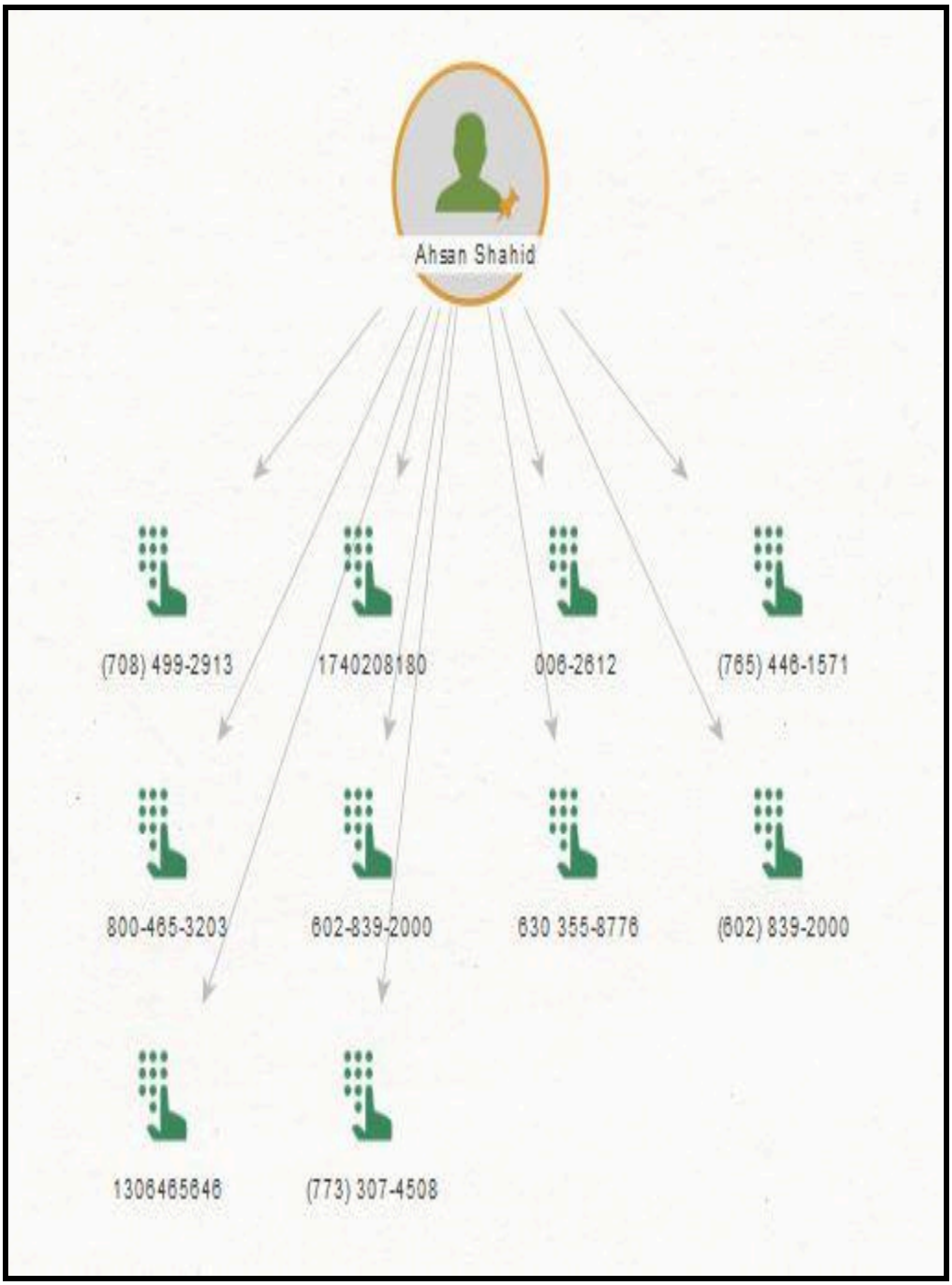
Target 2: Person

- Social Accounts



- Email addresses

- Phone numbers



7. Observations

- Maltego's transforms simplify complex data discovery
- Graphs provide immediate visual context for investigations
- Case studies revealed hidden infrastructure and contact trails
- Useful for identifying connections in both personal and corporate networks

8. Conclusion

Maltego proves to be a versatile and powerful OSINT tool that aids in digital investigations by visualizing relationships between data entities. Its capacity to automate data mining and reveal actionable intelligence makes it essential for security analysts, investigators, and researchers. When used correctly, it significantly reduces investigation time and increases result accuracy.